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$$\begin{cases} x' + 6x = 5 \\ x(0) = 4 \end{cases}$$

$$s\mathcal{L}[x] - x[4] + 6\mathcal{L}[x] = 5\mathcal{L}[1]$$

$$s\mathcal{L}[x] - 4 + 6\mathcal{L}[x] = \frac{5}{s}$$

$$(6+s)\mathcal{L}[x] = \frac{5}{s} + 4$$

$$\mathcal{L}[x] = \frac{5}{s(s+6)} + \frac{4}{6+s}$$

Fraccion Parcial

$$\frac{5}{s(s+6)} = \frac{A}{s} + \frac{B}{s+6}$$

$$5 = As + 6A + Bs$$

Para $s=$ $\quad A + B = 0 \cdots B = -\frac{5}{6}$

Para constante: $\quad 6A = 5 \cdots A = \frac{5}{6}$

$$\frac{5}{s(s+6)} = \frac{5}{6}\left[\frac{1}{s}\right] - \frac{5}{6}\left[\frac{1}{s+6}\right]$$

$$x = \frac{5}{6} - \frac{5}{6}e^{-6t} + 4e^{-6t}$$

$$= \frac{5}{6} + \frac{19}{6}e^{-6t}$$

$$= \frac{1}{6}\left[5 + 19e^{-6t}\right]$$

$$\begin{cases} 2x' + x = \sin(2t) \\ x(0) = 1 \end{cases}$$

$$2[s\mathcal{L}[x] - x[1]] + \mathcal{L}[x] = \frac{2}{s^2+4}$$

$$2s\mathcal{L}[x] - 2 + \mathcal{L}[x] = \frac{2}{s^2+4}$$

$$(2s+1)\mathcal{L}[x] = \frac{2}{s^2+4} + \frac{2(s^2+4)}{s^2+4}$$

$$(2s+1)\mathcal{L}[x] = \frac{2s^2+10}{s^2+4}$$

$$\mathcal{L}[x] = \frac{2s^2+10}{(s^2+4)(2s+1)}$$

Fracción Parcial

$$\frac{2s^2+10}{(s^2+4)(2s+1)} = \frac{As+B}{s^2+4} + \frac{C}{2s+1}$$

$$2s^2+10 = (As+B)(2s+1) + C(s^2+4)$$

$$2s^2+10 = 2As^2 + As + 2Bs + B + Cs + 4C$$

Para s^2 $\quad 2A + C = 2 \quad \cdots \quad C = 4B + 2 \quad \cdots = \frac{42}{17}$

Para s $\quad A + 2B = 0 \quad \cdots \quad A = -2B \quad = -\frac{16}{17}$

Para constante $\quad B + 4C = 10$

$$B + 16B + 8 = 10$$

$$17B + 8 = 10$$

$$17B = 2$$

$$B = \frac{2}{17}$$

$$\frac{2s^2+10}{(s^2+4)(2s+1)} = \frac{-\frac{8}{17} + \frac{2}{17}}{s^2+4} + \frac{42}{17} \cdot \frac{1}{2}\left[\frac{1}{s+1}\right]$$

$$x = \frac{1}{17}\left[\frac{-4s}{s^2+4} + \frac{2}{s^2+4} + \frac{21}{s+1}\right]$$

$$x = \frac{1}{17}\left[-4\cos(2t) + \sin(2t) + e^{-t}\right]$$

$$\begin{cases} x' + 2x = e^{-2t} \\ x(1) = 2 \end{cases}$$

$$s\mathcal{L}[x] - x[2] + 2\mathcal{L}[x] = \frac{1}{s+2}$$

$$s\mathcal{L}[x] - 2 + 2\mathcal{L}[x] = \frac{1}{s+2}$$

$$(s+2)\mathcal{L}[x] = \frac{1}{s+2} + \frac{2(s+2)}{s+2}$$

$$(s+2)\mathcal{L}[x] = \frac{2s+5}{s+2}$$

$$\mathcal{L}[x] = \frac{2s+5}{s^2+4s+4}$$

Fracción parcial

$$\frac{2s+5}{s^2+4s+4} \rightarrow \frac{As+B}{s^2+4s+4} = \frac{2s+5}{s^2+4s+4}$$

Para s $\quad A = 2$

Para constante $\quad B = 5$

$$\begin{cases} x'(t) = \sinh(3t) \\ x(0) = 2 \end{cases}$$

$$S\,\mathcal{L}[x] - x(0) + 2\,\mathcal{L}[1] = \mathcal{L}[\sinh(3t)]$$

$$S\,\mathcal{L}[x] - 2 + \frac{2}{S} = \frac{3}{s^2 - 9}$$

$$\mathcal{L}[x] = \frac{3}{S(S^2-9)} + \frac{2}{S} - \frac{2}{S^2}$$

Fracciones parciales

$$x = \frac{1}{3}(\cosh(3t) - 1) + 2t + 2$$

$$\begin{cases} x' + \frac{3}{2}x = t\,e^t \\ x(0) = 0 \end{cases}$$

$$S\,\mathcal{L}[x] - x(0) + \frac{3}{2}\mathcal{L}[x] = \mathcal{L}[t\,e^t]$$

$$\left(S + \tfrac{3}{2}\right)\mathcal{L}[x] = \mathcal{L}[t\,e^t]$$

$$\mathcal{L}[x] = \frac{1}{\left(S + \tfrac{3}{2}\right)(S-1)^2}$$

Fracciones parciales

$$\frac{1}{\left(S + \tfrac{3}{2}\right)(S-1)^2} = \frac{A}{S-1} + \frac{B}{(S-1)^2} + \frac{C}{\left(S + \tfrac{3}{2}\right)}$$

$1 = A(s-1)(s+\frac{3}{2}) + B(s+\frac{3}{2}) + C(s-1)^2$

$1 = As^2 + \frac{3}{2}As - \frac{3}{2}A + Bs + \frac{3}{2}B + Cs^2 - 2Cs + C$

Para constante: $-\frac{3}{2}A + \frac{3}{2}B + C = 1$

Para s^2: $A + C = 0 \quad \therefore \quad C = -A$

Para s: $\frac{1}{2}A + B - 2C = 0$

Entonces s $\quad \frac{1}{2}A + B + 2A = 0$

$B = -\frac{5}{2}A$

Entonces Para constate $\quad -\frac{3}{2}A + \frac{15}{4}A - A = 1$

$-\frac{25}{4}A = 1$

$A = -\frac{4}{25}$

$C = -A = \frac{4}{25} \qquad B = -\frac{5}{2}\left(-\frac{4}{25}\right) = \frac{2}{5}$

$x = \frac{4}{25}\left[\frac{1}{s+\frac{3}{2}}\right] - \frac{4}{25}\left[\frac{1}{s-1}\right] + \frac{2}{5}\left[\frac{1}{(s-1)^2}\right]$

$x = \frac{4}{25}e^{-\frac{3}{2}t} - \frac{4}{25}e^{t} + \frac{2}{5}te^{t}$